Fiscal Year:	FY 2007	**Task Last Updated:**	FY 06/03/2008
PI Name:	Collins, Lance R Ph.D.		
Project Title:	Investigation of Particle Clustering Using 3D Digital Holographic Imaging and Direct Numerical Simulations		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	09/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received no-cost extension through 9/30/2007, per PI (3/07)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Meng, Hui (SUNY Buffalo)		
Grant/Contract No.:	NNC05GA45G		
Performance Goal No.:			
Performance Goal Text:			
	It is proposed to develop digital holographic particle image velocimetry (DHPIV) that will make it possible to measure the three-dimensional position of approximately 10^5 particles with resolutions below 10 microns in all three directions. The instrument will be used to investigate particle clustering in turbulent flows. It is well known from direct numerical simulations (DNS) that particles, owing to their higher density, are 'centrifuged' out of regions of high vorticity and collect in regions of high strain. DNS further suggests this clustering can dramatically affect fundamental aerosol processes such as coagulation, vaporization/condensation and dispersion. However, there is limited experimental verification of the DNS observations to date, in large part due to the challenge of measuring particles in turbulence with sufficient accuracy to test the DNS. This proposal addresses this deficiency. In addition to the proposed cutting-edge experimental work, we will perform state-of-the-art DNS that %perfectly complement the experiments. The proposed		

Task Description:	DNS will use a code recently developed in our group that is capable of resolutions of 1024^3 on a 32-node 64-processor Beowulf cluster. We believe simulations with up to 2048^3 can be achieved on the SUNY-Buffalo Center for Computational Research (CCR) cluster, that has more than 300-nodes. The proposed simulations will be the largest multiphase simulations performed to date, and will enable closure of open questions concerning the Reynolds number dependence of particle clustering. The results have important implications for high-Reynolds-number applications such as astrophysics and atmospheric aerosols. In addition, the code has unique capabilities for simulating homogeneous shear flows, enabling us to explore the impact of this parameter on particle clustering, for the first time. The over-arching goal of this proposed study is to develop and validate a unique, next-generation experimental facility and DNS capability to investigate statistical properties of particles in turbulence. DHPIV has the potential to impact a broad range of experimental programs sponsored by NASA that would benefit from 3D particle position and velocity measurements from a single camera. Achieving this ambitious goal requires an interdisciplinary approach. The experimental facility and DHPIV will be developed by Prof. Meng, who has pioneered the use of holographic imaging in turbulence for the last ten years. The DNS will be performed by Prof. Collins, who recently has developed a new parallel algorithm that will enable unprecedented aerosol simulations. Each group brings complementary viewpoints and expertise that will benefit from the other, creating an exciting synergistic team to study turbulence. Any single investigator, quite simply, cannot accomplish the goals of this project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research involves the integration of direct numerical simulations and high resolution, three-dimensional holographic particle imaging. We are studying particle clustering in turbulence, a phenomenon thought to play an important role in the evolution of clouds on earth, and these in turn affect the weather and global climate change. This project is directed towards understanding the role of atmospheric turbulence on droplet clustering and cloud formation. Additionally, the project is developing more advanced simulation tools for particle-laden turbulence (in particular the extension to homogeneous turbulent shear flow). On the experimental side, we continue to push holographic imaging technology using digital cameras and our own image processing software that we continue to develop.
Task Progress:	The particle code has been coupled to our numerical Navier Stokes solver. In addition, gravitational settling of the particles has been incorporated. The code has been generalized to account for a particle size distribution and to store both Eulerian and Lagrangian statistics. Experimental conditions for the box turbulence experiment at SUNY-Buffalo were precisely matched in the direct numerical simulation (DNS) and comparisons of the radial distribution function (RDF) measured experimentally using holographic imaging and those obtained from the DNS were made. There is very good agreement between the two. The findings were published in the Journal of Fluid Mechanics.
Bibliography Type:	Description: (Last Updated: 10/05/2009)
Abstracts for Journals and Proceedings	Collins LR, Ayyalasomayajula S, Bodenschatz E, Cao L, Gylfason A, de Jong J, Meng H, Salazar JPLC, Warhaft Z, Woodward SH. "Experimental and numerical investigations of particle accelerations and clustering in isotropic turbulence. Invited talk." Multiphase Turbulence session, ASME 2nd Joint US–European Fluids Engineering Summer Meeting, Miami, FL, July 2006. ASME 2nd Joint US-European Fluids Engineering Summer Meeting, July 2006. , Jul-2006
Abstracts for Journals and Proceedings	Cao L, Woodward SH, Meng H, Salazar JPLC, Collins LR. "Study of aerosol particle clustering in isotropic turbulence using DNS and holographic PIV." Multiphase Particle Laden Flows III session, 58th Annual Meeting of the American Physical Society Division of Fluid Dynamics, Chicago, IL, November 2005. Bulletin of the American Physical Society Division of Fluid Dynamics 50:132, November 2005. , Nov-2005
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Articles in Peer-reviewed Journals	de Jong J, Cao L, Woodward SH, Salazar JPLC, Collins LR, Meng H. "Dissipation rate estimation from PIV in zero-mean-flow isotropic turbulence." Experiments in Fluids 2009 Mar;46(3):499-515. http://dx.doi.org/10.1007/s00348-008-0576-3 , Mar-2009
Articles in Peer-reviewed Journals	Brucker KA, Isaza JC, Vaithianathan T, Collins LR. "Efficient algorithm for simulating homogeneous turbulent shear flow without remeshing." Journal of Computational Physics 2007 Jul;225(1):20-32. http://dx.doi.org/10.1016/j.jcp.2006.10.018 , Jul-2007
Articles in Peer-reviewed Journals	Salazar JPLC, de Jong J, Cao L, Woodward SH, Meng H, Collins LR. "Experimental and numerical investigation of inertial particle clustering in isotropic turbulence." Journal of Fluid Mechanics. 2008 Apr;600:245-56. http://dx.doi.org/10.1017/S0022112008000372 , Apr-2008